



Sprint 2 Planning Document

Team 26: Xavion Marable, Logan Wilkinson, Emilio Guzman, Shreyansh Tehanguria, Shashank Mukkera

Sprint Overview

The goal of Sprint 2 is to expand the core of our application by implementing group coordination tools, safety features, and itinerary customization options. This sprint focuses on moving past the basic authentication into the “live” trip experience, with real-time alerts, group role management, and financial coordination. While Sprint 1 was about getting users into the system, this sprint is about what happens when they actually start their trip. We are shifting more towards the front end this time. We realized in the last sprint that a functional backend doesn’t mean much if the interface is cluttered or confusing, so we are going to prioritize a clean, high-contrast, responsive UI. We want the app to feel intuitive whether a user is trying to find a place to put their bags or in a high-stress situation needing emergency numbers. This means building better navigation, smarter state management, and a more polished look for everything.

Scrum Master: Xavion Marable

Meeting Schedule: Tuesdays, Wednesdays, Thursdays @ 4:30 PM (Discord)

Risks and Challenges

- **External API Integration** - Features like Bag Storage and the SOS button rely on external data. If the API is slow or returns weirdly formatted data, it could break out layouts. We will need to build high quality loading states and error skeletons so the UI doesn’t just crash.
- **UI Polish** - Since we are focusing much more on the UI, the biggest risk is overcomplicating the components. We need to keep CSS modular and all of the components small, which will be a big challenge.
- **Role-Based Access Control (RBAC)** - Implementing Role Assignment means we have to be super careful with how we show or hide buttons. We do not want a Viewer to see an Edit button that just throws an error when it is clicked. The UI should adapt to their role.
- **Concurrency in Groups** - With the group board and rainy day plans, we have to make sure that when one person pins a note or swaps the schedule, everyone else sees it without having to manually refresh the page.

2. Current Sprint Detail

User Story #1: SOS Button

"As a user, I would like to be able to access an SOS button that shows emergency numbers."

- **Owner:** Xavion Marable
- **Estimated Hours:** 4.5 Hours
- **Tasks:**
 - Create a localized database or API integration for international emergency numbers (Police, Fire, Ambulance). (1.5 hrs)
 - Build the SOS overlay UI with high-contrast buttons and one-tap dialing functionality. (1 hr)
 - Implement offline caching to ensure numbers are visible without an active data connection. (0.5 hr)
 - Write unit and integration tests to verify localized number fetching based on GPS coordinates and offline availability of cached emergency data. (1.5 hrs)
- **Acceptance Criteria:**
 - Given I am in a foreign country, When I tap the SOS button, Then the app displays the specific local emergency services numbers for that region.
 - Given I am in a high-stress situation, When I open the SOS menu, Then the interface must use large touch targets to prevent accidental misclicks.
 - Given I have no cellular data, When I trigger the SOS view, Then the numbers remain accessible via offline storage.

User Story #2: Group Board

"As a user, I would like to be able to view a “Group Board” for pinned announcements."

- **Owner:** Xavion Marable
- **Estimated Hours:** 5 Hours
- **Tasks:**

- Extend the TravelGroup schema to include a pinnedAnnouncements array. (1 hr)
- Build the Group Board UI to display messages with "pinned by" metadata and timestamps. (1.5 hrs)
- Create the logic for leaders to add or remove messages from the board. (1 hr)
- Write unit and integration tests to verify pinned message persistence in the database and real-time updates for all group members via socket or polling. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** a group leader pins a message, **When** I navigate to the Group Board, **Then** that message appears at the top of the board.
 - **Given** there are multiple announcements, **When** I view the board, **Then** I can see exactly which leader pinned each update.
 - **Given** an announcement is unpinned by a leader, **When** I refresh the board, **Then** the item is removed from the view for all members.

User Story #3: Role Assignment

"As a user, I would like to be able to assign roles (Admin, Viewer) to group members."

- **Owner:** Xavion Marable
- **Estimated Hours:** 6 Hours
- **Tasks:**
 - Update the User/Group junction model to include a role field (Leader, Admin, Viewer). (1.5 hrs)
 - Create a Role Management UI accessible only to the Group Leader. (2 hrs)
 - Implement middleware to restrict specific API routes based on the user's role. (1 hr)

- Write unit and integration tests to verify role-based access control (RBAC) and ensure "Viewer" accounts are blocked from edit-only API routes. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** I am the group leader, **When** I open the member list, **Then** I can toggle a user between 'Admin' and 'Viewer' status.
 - **Given** a member is a 'Viewer', **When** they attempt to edit the itinerary, **Then** the UI disables all save buttons and the API blocks the request.
 - **Given** the group needs a new primary contact, **When** I transfer 'Leader' status to another member, **Then** my own permissions are downgraded and the new leader gains full control.

User Story #4: External Booking

"As a user, I would like to be able to go to activity booking sites via external links."

- **Owner:** Xavion Marable
- **Estimated Hours:** 4.5 Hours
- **Tasks:**
 - Add an externalBookingUrl field to the Activity schema. (0.5 hr)
 - Build the redirection logic with a security warning for external sites. (1.5 hrs)
 - Implement UI logic to hide the booking button if no URL is provided by the API. (1 hr)
 - Write unit and integration tests to verify external link sanitization and ensure the booking button visibility logic matches API availability. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** I am looking at a museum in the itinerary, **When** I click "Book Now," **Then** the app opens the vendor's site in a new browser tab.
 - **Given** an activity does not require a reservation, **When** I view its details, **Then** the booking button is hidden to avoid confusion.

- **Given** a booking link is missing, **When** the itinerary is generated, **Then** the app provides a generic "Search for Tickets" link as a fallback.

User Story #5: Vibe Tags

"As a user, I would like to be able to select specific “Vibe” tags (Nightlife, Nature, Historical) to guide the itinerary generator."

- **Owner:** Xavion Marable
- **Estimated Hours:** 6.5 Hours
- **Tasks:**
 - Design the Vibe Tag selector UI with multi-select capabilities. (2 hrs)
 - Update the itinerary generator logic to prioritize locations matching the selected tags. (2 hrs)
 - Implement a "Reset to Balanced" function to clear all active tags. (1 hr)
 - Write unit and integration tests to verify that the itinerary generator correctly filters activities based on multiple concurrent vibe tags. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** I select the "Nature" and "Historical" tags, **When** I generate a trip, **Then** the system excludes clubs in favor of parks and landmarks.
 - **Given** I change tags mid-planning, **When** I regenerate the plan, **Then** the new itinerary reflects the updated vibes without changing "locked" items.
 - **Given** no tags are selected, **When** I hit generate, **Then** the system provides a balanced "General Interest" itinerary by default.

User Story #6: Bag Storage

"As a user, I would like to be able to find temporary bag storage."

- **Owner:** Xavion Marable
- **Estimated Hours:** 5 Hours

- **Tasks:**
 - Integrate a third-party API to fetch storage locations. (2 hrs)
 - Build the Storage Search UI with a list/map toggle view. (1 hr)
 - Add detail cards showing pricing and hours for selected shops. (0.5 hr)
 - Write unit and integration tests to verify external API data mapping and distance-based filtering for storage locations. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** I have a late flight after checkout, **When** I search for "bag storage" near me, **Then** the app shows a list of verified nearby lockers.
 - **Given** I select a storage location, **When** I tap it, **Then** the app displays the pricing, operating hours, and a "Navigate" button.
 - **Given** I am in a city with no partner options, **When** I search, **Then** the app indicates if no verified lockers are available within a 5-mile radius.

User Story #7: Travel Reminders

"As a user, I would like to be able to create reminders for my travels."

- **Owner:** Xavion Marable
- **Estimated Hours:** 6 Hours
- **Tasks:**
 - Build a personal "Trip Checklist" UI for manual reminder entries. (2 hrs)
 - Implement push notification triggers using a background task scheduler. (1.5 hrs)
 - Link reminders to the shared calendar so changes in event times auto-update the reminder time. (1 hr)
 - Write unit and integration tests to verify that notification triggers automatically adjust when the linked calendar event time is updated. (1.5 hrs)
- **Acceptance Criteria:**

- **Given** I have a flight, **When** I set a reminder for "3 hours before," **Then** I receive a push notification even if the app is closed.
- **Given** I add a task like "Buy sunscreen," **When** I view my trip reminders, **Then** it appears in a personal checklist.
- **Given** my flight time changes in the shared calendar, **When** the event updates, **Then** any linked reminders automatically shift to the new time.

User Story #8: Rainy Day Plans

"As a user, I would like to generate a "Rainy Day" alternative for my plans."

- **Owner:** Xavion Marable
- **Estimated Hours:** 7.5 Hours
- **Tasks:**
 - Create a "Backup Plan" generator logic that filters for indoor-only categories. (2.5 hrs)
 - Build the UI toggle to switch between the "Primary" and "Rainy Day" itinerary views. (2 hrs)
 - Implement a side-by-side comparison view to help groups vote on the switch. (1.5 hrs)
 - Write unit and integration tests to verify the "indoor-only" filtering logic and the side-by-side comparison state management. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** it starts raining, **When** I hit the "Rainy Day" toggle, **Then** the app replaces outdoor spots with indoor alternatives for the same time slots.
 - **Given** I want to be prepared, **When** I generate my initial plan, **Then** the app automatically creates a "Backup Itinerary."
 - **Given** no indoor alternative exists for a specific activity, **When** I toggle the view, **Then** the app suggests a generic fallback like a "Coffee Shop."

User Story #9 Split Costs:

"As a user, I would like to be able to split the costs with my group."

- **Owner:** Emilio Guzman
- **Estimated Hours:** 16.5 hrs
- **Tasks:**
 - Create cost-splitting schema in Mongoose to support equal and custom splits for a shared expense (groupId, tripId, expenseId, participants, splitType, amounts, createdBy, createdAt, updatedAt) (2 hrs)
 - Build helper logic to calculate each member's share based on equal split or custom amounts/percentages (2 hrs)
 - Build POST `/api/groups/:groupId/cost-splits` to create a split for an expense with validation (group membership, valid participants, total split matches expense amount) (2 hrs)
 - Build GET `/api/groups/:groupId/cost-splits` to fetch group cost splits and filter by trip or expense (1.5 hrs)
 - Build PUT `/api/groups/:groupId/cost-splits/:id` to edit a cost split (only creator or group admin can edit) (2 hrs)
 - Build DELETE `/api/groups/:groupId/cost-splits/:id` to remove a cost split (permission check) (1 hr)
 - Create Split Costs UI section (expense selection, participant selector, equal/custom split options, amount preview) (3 hrs)
 - Connect UI → API (create, update, delete split, optimistic updates, show validation errors) (1.5 hrs)
 - Write unit and integration tests to verify create, read, update, and delete functionality for cost splits. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** I am a member of a travel group, **When** I create a valid split for a shared expense, **Then** the split is saved and each participant's share is recorded correctly.

- **Given** I choose an equal split, **When** I submit the expense split, **Then** the total cost is divided evenly among the selected group members.
- **Given** I choose a custom split, **When** I submit participant-specific amounts or percentages, **Then** the system saves the split only if the total matches the expense amount.
- **Given** I submit a split with invalid participant data or totals that do not match the expense amount, **When** I click save, **Then** the form blocks submission and shows an error.
- **Given** I am not a member of the group, **When** I request the cost splits endpoint, **Then** the API denies access (401/403).
- **Given** a cost split exists, **When** I edit or delete it as the creator or group admin, **Then** the change is persisted and visible to the group.
- **Given** I am not the creator of a cost split and not an admin, **When** I attempt to edit or delete the split, **Then** the system denies access and returns a 403 error.

User Story #10 Post Shared cost:

"As a user, I would like to be able to post a shared cost to the group ledger."

- **Owner:** Emilio Guzman
- **Estimated Hours:** 16.5 hrs
- **Tasks:**
 - Create SharedCost schema in Mongoose (groupId, tripId, title, description, amount, currency, category, paidBy, participants, splitType, date, notes, createdBy, createdAt, updatedAt) (2 hrs)
 - Build POST `/api/groups/:groupId/shared-costs` to create a shared cost entry with validation (required fields, positive amount, valid payer, valid group participants) (2 hrs)

- Build GET `/api/groups/:groupId/shared-costs` to fetch shared costs with filters by trip, category, payer, or date (1.5 hrs)
- Build PUT `/api/groups/:groupId/shared-costs/:id` to edit a shared cost (only creator or group admin can edit) (2 hrs)
- Build DELETE `/api/groups/:groupId/shared-costs/:id` to remove a shared cost entry (permission check) (1 hr)
- Build backend integration so posting a shared cost can automatically generate or update a related cost split or ledger entry (2 hrs)
- Create Shared Costs UI page (cost list, add modal, payer selector, participant selector, category/date inputs) (3 hrs)
- Connect UI → API (create, update, delete, optimistic updates, show validation and save errors) (1.5 hrs)
- Write unit and integration tests to verify create, read, update, and delete functionality for shared costs, including validation rules, ledger updates, and proper authorization for group membership and permissions. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** I am a member of a travel group, **When** I post a valid shared cost to the group ledger, **Then** the cost is saved and visible in the group ledger.
 - **Given** I enter all required information for a shared cost, **When** I click save, **Then** the system records the amount, payer, participants, and other details correctly.
 - **Given** I submit a shared cost with missing required fields or an invalid amount, **When** I click save, **Then** the form blocks submission and shows an error.
 - **Given** I am not a member of the group, **When** I call the shared costs API endpoints, **Then** access is denied (401/403).
 - **Given** a shared cost exists, **When** I edit or delete it as the creator or group admin, **Then** the updated ledger entry is persisted and visible to the group.

- **Given** I am not the creator of a shared cost and not an admin, **When** I attempt to edit or delete the entry, **Then** the system denies access and returns a 403 error.
- **Given** a shared cost is posted, **When** the ledger is recalculated, **Then** the corresponding balances reflect the new expense correctly.

User Story #11 Group Photo Album:

“As a user, I would like to be able to create a shared photo album for trips.”

- **Owner:** Logan Wilkinson
- **Estimated Hours:**
- **Tasks:** 12.5 hrs
 - Build the **GET** `/api/group/[groupId]/album` (2 hrs)
 - Build the **POST** `/api/group/[groupId]/album`. (3 hrs)
 - Build the **DELETE** `/api/group/[groupId]/album` (2 hrs)
 - Make the Upload a Photo UI. (2 hrs)
 - Make the Group Photo Album UI. (2 hrs)
 - Write unit and integration tests to ensure photos are uploaded, fetched, and deleted correctly while maintaining group-level access. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** that I'm on the Upload a Photo page, **When** I upload an image(s) and submit, **Then** the user will be redirected back to the album page.

- **Given** that I'm on the Photo Album page, **When** I delete a photo, **Then** the change will take effect and be reflected in the group immediately.
- **Given** that I visit my groups photo album page, **When** it loads, **Then** the page should show all uploaded photos.
- **Given** that I am on the Upload a Photo page, **When** I upload an image that is too large, **Then** the page should show an "Image too large" error.
- **Given** that I am on the Upload a Photo page, **When** I submit no photos, **Then** the page should show an "No images given" error.

User Story #12 Manage Push Notifications:

"As a user, I would like to be able to manage my notification preferences (Email, SMS, Push) for trip updates."

- **Owner:** Logan Wilkinson
- **Estimated Hours:** 5.5 hours
- **Tasks:**
 - Add notification settings to **User** schema. (0.5 hr)
 - Implement updating of new user settings in `/api/user/me` PATCH and GET routes. (2.5 hrs)
 - Add new settings to user settings page. (1 hr)
 - Write unit and integration tests to verify notification preference updates and retrieval. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** that I'm on the account settings page, **When** I change any settings, **Then** the change will take effect and be reflected on the account immediately.
 - **Given** I am logged in, **When** I edit my account settings, **Then** the change appears on my account immediately.

- **Given** I disable push notifications, **When** a trip update occurs, **Then** I do not receive a push notification on my device.

User Story #13 Group Email Notification:

“As a user, I would like to be able to send text notifications to my group.”

- **Owner:** Logan Wilkinson
- **Estimated Hours:** 6.5 hours
- **Tasks:**
 - Create `POST /api/group/[groupId]/sms` route. (3 hr)
 - Create Send Text Notification component and integrate into group page (2 hr)
 - Write unit and integration tests to verify user-level notification opt-out enforcement. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** that I'm on the group page, **When** I submit a message to send via sms to my group, **Then** the message will be sent out.
 - **Given** that I'm on the group page, **When** I submit a message to send via sms to my group, **Then** I will get either a success or failure message.
 - **Given** I am on the group page, **When** I try to submit an empty text, **Then** I will get an error message.

User Story #14 Activity Preferences (Avoid + Budget Suggestions)

“As a user, I would like to mark activities or locations to avoid and receive travel recommendations based on my budget.”

- **Owner:** Shreyansh Tehanguria
- **Estimated Hours:** 11 hrs
- **Tasks:** 10 hrs
 - Update the Trip/TripSettings schema to include an **avoidActivities** or **avoidLocations** field to store activities users want to exclude. **(1 hrs)**
 - Implement backend filtering logic so the itinerary generator excludes avoided activities when generating recommendations. **(2 hrs)**
 - Build the budget-based location recommendation logic that suggests destinations or activities based on the user's selected budget range. **(2.5 hrs)**
 - Create API routes such as **POST /api/trips/preferences** and **GET /api/trips/budget-recommendations** to store and retrieve user preferences. **(1.5 hrs)**
 - Build the UI for selecting avoided activities and setting a budget range using dropdowns and multi-select components. **(2 hrs)**
 - Connect the UI to the API so preferences update and regenerate the itinerary when changed. **(1 hr)**
 - Write unit and integration tests to verify budget-range filtering and activity exclusion logic. **(1.5 hrs)**
- **Acceptance Criteria:**
 - **Given** I select activities or locations to avoid, **When** I generate an itinerary, **then** those activities should not appear in the generated results.
 - **Given** I enter a trip budget, **When** I request suggestions, **Then** the system provides recommendations generally within that budget range.
 - **Given** I update my preferences, **When** I regenerate the itinerary, **then** the updated preferences should be reflected in the results.

User Story #15 Activity Information Page

“As a user, I would like to view additional information about activities so I can learn more before adding them to my itinerary.”

- **Owner:** Shreyansh Tehanguria
- **Estimated Hours:** 10.5 hrs
- **Tasks:**
 - Extend the Activity schema to include fields such as `infoUrl`, `description`, and `referenceLinks` for additional information sources. **(2 hrs)**
 - Build the **GET /api/activities/[activityId]** API route to fetch detailed activity information. **(1.5 hrs)**
 - Create the Activity Information UI page displaying description, location details, and informational links. **(2.5 hrs)**
 - Implement UI logic that dynamically renders information links for each activity. **(1.5 hrs)**
 - Add loading states and fallback UI in case activity details are unavailable or slow to load. **(1 hr)**
 - Connect itinerary activity cards to the new Activity Information page. **(1 hr)**
 - Write unit and integration tests to verify deep-linking to activity IDs and data fetching accuracy. **(1.5 hrs)**
- **Acceptance Criteria:**
 - **Given** I click on an activity from the itinerary, **When** the activity page opens, **then** I should see more information about that activity.
 - **Given** the activity contains additional information links, **When** I click them, **then** they should open in a new tab.
 - **Given** an activity does not have external links, **When** I open the page, **then** the activity details should still display without errors.

User Story #16 External Booking Links

“As a user, I would like to access external booking links for activities so I can reserve tickets or make bookings easily.”

- **Owner:** Shreyansh Tehanguria
- **Estimated Hours:** 10 hrs
- **Tasks:** 10 hrs
 - Update the Activity schema to include an optional `bookingUrl` field for external ticket booking vendors. **(2 hrs)**
 - Build backend logic so the booking link is returned when fetching activity details. **(1.5 hrs)**
 - Implement the **Book Now** button component that redirects users to the external vendor site. **(2 hrs)**
 - Add a security prompt or visual indicator informing users that they are leaving the app for an external website. **(1.5 hrs)**
 - Implement UI logic to hide the booking button if the activity does not have a booking link. **(1.5 hrs)**
 - Write a unit test to verify correct behavior for activities with and without booking links. **(1.5 hrs)**
- **Acceptance Criteria:**
 - **Given** an activity has a booking URL, **When** I click the **Book Now** button, **then** the vendor’s site should open in a new tab.
 - **Given** an activity does not require booking, **When** I view its details, **then** the booking button should not appear.
 - **Given** I open a booking link, **When** the new tab loads, **Then** the main application should continue working normally.

User Story #17 Group Polls:

“As a user I would like to be able to create “Polls” within the group to vote on destination options.”

- **Owner:** Logan Wilkinson
- **Estimated Hours:** 9.5 hours
- **Tasks:**

- Create **POST** `/api/group/[groupId]/poll` route. (2 hr)
- Create **GET** `/api/group/[groupId]/poll` route. (2 hr)
- Create **DELETE** `/api/group/[groupId]/poll` route. (2 hr)
- Create Poll component and integrate into group page (2 hr)
- Write a unit test to make sure polls are created and updated successfully. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** that I'm on the group page, **When** a poll is active, **Then** the poll will be present on the page.
 - **Given** I am on the group page, **When** I vote on a poll, **Then** the change appears immediately on my end.
 - **Given** I am on the group page, **When** I remove or change my vote on a poll, **Then** the change appears immediately on my end.
 - **Given** I am on the group page, **When** I click delete on a poll I created, **Then** the poll is deleted from the group and change appears immediately.
 - **Given** I am on the group page, **When** I am viewing the groups polls, **Then** there is no delete button for polls I did not create.

User Story #18: View Trip Expense Summary

"As a user, I would like to be able to view a list of what everyone owes at the end of the trip."

- **Owner:** Shashank Mukkera
- **Estimated Hours:** 8 Hours
- **Tasks:**
 - Build GET /api/groups/{id}/ledger/summary to compute net balances from the existing **ledger** (Expense array) on the TravelGroup document: iterate each expense's **debtors** map, aggregate per-member totals, and return a simplified list of "User A owes User B \$X" settlements using a min-cash-flow algorithm. (2.5 hrs)
 - Create the "Expense Summary" UI page/tab within the group view: display a table/card list showing each member's net balance (positive = owed money, negative = owes money) and the optimized settlement plan ("A pays B \$X"). (2 hrs)
 - Add filtering/sorting controls on the summary UI: filter by settled vs. unsettled expenses, sort settlements by amount, and show a breakdown toggle to view individual expenses behind each balance. (1.5 hrs)
 - Handle edge cases in the API: empty ledger returns zeroes, single-member group shows no debts, all-settled ledger shows "All settled up!" message. Ensure requester is a group member (403 otherwise). (1 hr)
 - Write unit tests: net balances computed correctly for multiple overlapping expenses; min-cash-flow reduces transaction count; settled expenses excluded from active summary; non-members receive 403. (1 hr)
- **Acceptance Criteria:**
 - Given that I am a member of a group with recorded expenses, When I navigate to the Expense Summary page, Then I see each member's net balance and an optimized list of who owes whom and how much.
 - Given that all expenses in the group are marked as settled, When I view the summary, Then I see a message indicating all debts are settled.

- **Given that I am not a member of the group, When I request GET /api/groups/{id}/ledger/summary, Then I receive 403 Forbidden.**
- Given that the group has no expenses, When I view the summary, Then I see a message indicating there are no expenses to summarize.

User Story #19: Request Payment from Group Members

"As a user, I would like to be able to request payment from specific group members for a shared cost."

- **Owner:** Shashank Mukkera
- **Estimated Hours:** 9.5 Hours
- **Tasks:**
 - Add a PaymentRequest sub-schema to the TravelGroup model (requestID, requesterID, targetMemberID, amount, expenseID reference, status: "pending" | "paid" | "declined", createdAt, message). (1.5 hrs)
 - Build POST /api/groups/{id}/payment-requests to create a new payment request: validate the requester is a group member, the target is a group member, amount is positive and does not exceed the computed debt, and link it to the relevant expense(s). (2 hrs)
 - Build GET /api/groups/{id}/payment-requests to list all payment requests for the group (filterable by status and by user — "requests I sent" vs. "requests I received"). (1.5 hrs)
 - Build PATCH /api/groups/{id}/payment-requests/{requestId} to allow the target member to mark a request as "declined" with an optional reason. (1 hr)
 - Create the "Request Payment" UI: a button on the expense summary or member balance card that opens a modal pre-filled with the owed amount and target member, with Zod-validated fields (amount, optional message), and a send button. (2 hrs)

- Create the "My Payment Requests" UI section: show incoming requests with "Pay" and "Decline" actions, and outgoing requests with status badges (pending/paid/declined). (1.5 hrs)
- Write unit tests: payment request created with correct fields; amount exceeding debt is rejected; non-members cannot create requests (403); target can decline; duplicate pending request for same expense is prevented. (1.5 hrs)
- **Acceptance Criteria:**
 - Given that I am a group member and another member owes me money, When I select that member and submit a payment request with a valid amount, Then a payment request is created with status "pending" and the target member can see it in their incoming requests.
 - Given that I submit a payment request with an amount exceeding what the member owes, When I hit submit, Then the form shows an error ("Amount exceeds outstanding balance") and does not submit.
 - Given that I receive a payment request, When I click "Decline," Then the request status changes to "declined" and the requester sees the updated status.
 - Given that I am not a member of the group, When I try to create a payment request, Then the request is rejected (403).

User Story #20: Send Payment Confirmation Notice

"As a user, I would like to be able to send a payment confirmation notice to a member after paying them back."

- **Owner:** Shashank Mukkera
- **Estimated Hours:** 7.5 Hours
- **Tasks:**
 - Build POST
/api/groups/{id}/payment-requests/{requestId}/confirm to allow the requester to confirm payment: update the payment request

status to "paid", record a confirmation timestamp, and mark the linked expense's debtor entry as settled if fully paid. (2 hrs)

- Build a notification mechanism: when a payment is confirmed, create an in-app notification for the requester (add a Notification sub-schema or collection: notificationID, recipientID, type: "payment_confirmed", groupID, paymentRequestID, read: boolean, createdAt). (1.5 hrs)
- Build GET /api/notifications (or GET /api/user/me/notifications) to return the current user's notifications (paginated, sorted by createdAt descending) and PATCH /api/notifications/{id}/read to mark as read. (1.5 hrs)
- Create the "Confirm Payment" UI flow: on an incoming payment request card, show a "Confirm Payment" button that opens a confirmation dialog ("Confirm you have paid \$X to [Name]?"), then sends the API call and shows a success toast. (1 hr)
- Create a notification indicator (bell icon with unread count in the navbar) and a dropdown/page showing recent notifications with links to the relevant group/payment request. (1.5 hrs)
- Write unit tests: confirming payment updates request status to "paid" and creates a notification; only the target member can confirm; confirming an already-paid request returns 400; notification is marked as read via PATCH; expense debtor is marked settled when fully paid. (1.5 hrs)

- **Acceptance Criteria:**

- Given that I have a pending payment request where I am the requester, When I click "Confirm Payment" and confirm the dialog, Then the request status updates to "paid" and the payer receives an in-app notification.
- Given that a payment I requested has been confirmed, When I check my notifications, Then I see a notification stating "[Name] has confirmed payment of \$X" with a link to the group.
- Given that a payment request is already marked as "paid," When I try to confirm it again, Then I receive a 400 error ("Payment already confirmed").

- Given that the confirmed payment fully covers my debt for that expense, When the confirmation is processed, Then my debtor entry on that expense is marked as settled in the ledger.

User Story #21: Regenerate Parts of the Itinerary

"As a user, I would like to be able to click a button to regenerate specific parts of the itinerary."

- **Owner:** Shashank Mukkera
- **Estimated Hours:** 9 Hours
- **Tasks:**
 - Build POST `/api/groups/{id}/itinerary/regenerate` that accepts a list of CalendarEvent IDs (or a date range / category filter) to regenerate: fetch the group's MustHave items and trip details, call the AI/LLM itinerary generation service for only the selected portion, and return the newly generated events. (2.5 hrs)
 - Implement a "preview before apply" flow in the API: the regenerate endpoint returns proposed new events without saving; a separate POST `/api/groups/{id}/itinerary/regenerate/apply` accepts the proposed events and replaces the old ones (delete old CalendarEvents with matching IDs, insert new ones). (2 hrs)
 - Add selection UI to the itinerary view: checkboxes or a lasso-select on individual CalendarEvent cards/timeline blocks, plus a "Select by day" option, with a "Regenerate Selected" button that becomes active when at least one event is selected. (2 hrs)
 - Create the regeneration preview modal: show the proposed new events side-by-side with the originals (before/after), with "Accept" and "Cancel" buttons. On accept, call the apply endpoint and refresh the itinerary view. (1.5 hrs)
 - Write unit tests: only selected events are regenerated (others untouched); regeneration respects MustHave items marked as "approved"; non-members receive 403; applying proposed

events correctly replaces old events in the database; cancelling preview does not alter existing events. (1 hr)

- **Acceptance Criteria:**

- Given that I am a group member viewing the itinerary, When I select one or more events and click "Regenerate Selected," Then I see a preview of the newly generated events alongside the originals.
- Given that I am viewing the regeneration preview, When I click "Accept," Then the selected events are replaced with the new ones and the itinerary view updates accordingly.
- Given that I am viewing the regeneration preview, When I click "Cancel," Then no changes are made to the itinerary and I return to the original view.
- Given that the group has approved MustHave items, When I regenerate a portion of the itinerary, Then the regenerated events still incorporate the approved must-haves relevant to that portion.
- Given that I am not a member of the group, When I try to regenerate, Then I receive 403 Forbidden.

3. Backlog (Future Sprints)

User Account

- ~~As a user, I would like to be able to create an account.~~
- ~~As a user, I would like to be able to login and manage my account settings.~~
- ~~As a user, I would like to be able to edit my profile details, like name, school, location, and profile picture.~~
- ~~As a user, I would like to be able to reset my password via email/sms if I forget it.~~
- As a user, I would like to be able to link my external calendar (Google/Outlook) to sync trip dates.

- ~~As a user, I would like to be able to verify my student status (.edu email) to unlock student filters.~~
- ~~As a user, I would like to be able to manage my notification preferences (Email, SMS, Push) for trip updates.~~

Travel Groups

- ~~As a user, I would like to be able to create travel groups.~~
- ~~As a user, I would like to be able to manage my travel group.~~
- ~~As a user, I would like to be able to leave my travel group.~~
- ~~As a user, I would like to be able to access a shared calendar for my travel group.~~
- ~~As a group leader, I would like to be able to transfer my leader's permission to another member.~~
- As a group leader, I would like to be able to remove specific people from the travel group.
- ~~As a user, I would like to be able to message the members of my travel group.~~

Trip Creation

- ~~As a user, I would like to be able to input my travel preferences.~~
- ~~As a user, I would like to create a budget for the trip.~~
- ~~As a user, I would like to be able to specify transportation preferences.~~
- As a user, I would like to be able to add activities/locations as must haves.
- ~~As a user, I would like to be able to add activities/locations to avoid.~~
- ~~As a user, I would like to be able to generate a random location based on budget.~~
- As a user, I would like to be able to see recommendations/extra info for activities.

Itinerary Interaction/Modification

- ~~● As a user, I would like to be able to see reviews for places people have travelled.~~
- ~~● As a user, I would like to be able to see a summary of reviews for activities.~~
- ~~● As a user, I would like to be able to visit sites for more information about activities.~~
- ~~● As a user, I would like to be able to go to activity booking sites via external links.~~
- ~~● As a user, I would like to be able to find temporary bag storage.~~
- As a user, I would like to be able to have an offline itinerary view.
- As a user, I would like to be able to see a visual map representation of the itinerary.
- As a user, I would like to be able to have an offline map view.
- ~~● As a user, I would like to be able to create reminders for my travels.~~
- ~~● As a user, I would like to be able to access an SOS button that shows emergency numbers.~~
- ~~● As a user, I would like to be able to write reviews for places I've visited.~~
- ~~● As a user I would like to be able to create "Polls" within the group to vote on destination options.~~
- ~~● As a user, I would like to be able to assign roles (Admin, Viewer) to group members.~~
- ~~● As a user, I would like to be able to view a "Group Board" for pinned announcements.~~
- As a user, I would like to be able to generate alternative plans.
- As a user, I would like to be able to make changes to only one part of an itinerary.
- As a user, I would like to be able to click a button to regenerate specific parts of the itinerary.
- As a user, I would like to be able to click a button to remove options.
- As a user, I would like to be able to vote on specific activity options created by the itinerary generator.
- As a user, I would like to be able to retrieve previous itineraries.
- As a user, I would like to be able to post itineraries visible to all app users.

- ~~As a user, I would like to be able to share itineraries via link.~~
- As a user, I would like to be able to export itineraries to my preferred calendar.
- ~~As a user, I would like to be able to select specific “Vibe” tags (Nightlife, Nature, Historical) to guide the itinerary generator.~~
- As a user, I would like to be able to lock specific activities I like so they don’t change during regeneration.
- As a user, I would like to be able to drag and drop activities to reorder the schedule visually.
- ~~As a user, I would like to generate a “Rainy Day” alternative for my plans.~~
- As a user, I would like to be able to filter generated locations based on any accessibility requirements.
- ~~As a user, I would like to be able to view a 10-day weather forecast in the itinerary.~~

Itinerary Budget

- ~~As a user, I would like to be able to post a shared cost to the group ledger.~~
- ~~As a user, I would like to be able to split the costs with my group.~~
- ~~As a user, I would like to be able to view a list of what everyone owes at the end of the trip.~~
- ~~As a user, I would like to be able to request payment from specific group members for a shared cost.~~
- ~~As a user, I would like to be able to send a payment confirmation notice to a member after paying them back.~~

Communication

- ~~As a user, I would like to be able to connect with new people by finding and adding them as friends.~~
- ~~As a user, I would like to be able to create a shared photo album for trips.~~
- ~~As a user, I would like to be able to send text notifications to my group.~~

- ~~● As a user, I would like to be able to send email notifications to my group.~~
- As a user, I would like to be able to send in-app notifications to my group.
- As a user, I would like to be able to receive text notifications about my travel.
- As a user, I would like to be able to receive email notifications about my travel.
- ~~● As a user, I would like to be able to receive in-app notifications about my travel.~~
- As a user, I would like to be able to share my real-time location with added friends.